AVOCADO AGENCY

Here is where your presentation begins

Today's AGENDA

Midterm Logistics

Proctoring guidelines

Formatting instructions

Question 1: Spring 2023 Midterm Exam 1 - Problem 4

Question 2

Question 3 if time

TABLE OF CONTENTS

01

Exam Logistics

Proctoring

Formatting

Midterm Logistics

Format

90 minutes, on paper, no calculators or electronics except viewing the exam and proctoring

Content

Lectures 1-5, Homeworks 1-4, Groupworks 1-2

When and Where

Wednesday, July 15th,
11:00am-12:48pm PT
Zoom

02

Spring 2023

Midterm Exam 1

Problem 4

Problem 4

Suppose we are given a dataset of points $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$ and for some reason, we want to make predictions using a prediction rule of the form

$$H(x) = \underbrace{17}_{w_0} + w_1 x.$$

Prediction rule: $H(x) = 17 + w_1 x$

Problem 4.1

Write down an expression for the mean squared error of a prediction rule of this form, as a function of the parameter w_1.

$$R_{sq}(H(x)) = \frac{1}{n}\sum_{i=1}^{n}(y_i - H(x_i))^2$$

$$= \frac{1}{n}\sum_{i=1}^{n}(y - (17 + w_1 x_i))^2$$

Prediction rule: $H(x) = 17 + w_1 x$

Problem 4.2

Minimize the function $MSE(w_1)$ to find the parameter w_1^* which defines the optimal prediction rule $H^*(x) = 17 + w_1^* x$. Show all your work and explain your steps.

* derivative = 0 solve w_1

$$R_{sq}(H(x)) = \frac{1}{n}\sum_{i=1}^{n}(y_i - 17 - w_1 x_i)^2$$

$$0 = \frac{1}{n}\sum_{i=1}^{n} -2x_i(y_i - 17 - w_1 x_i)$$

Chain rule

$$0 = \frac{1}{n}\sum -2x_i(y_i - 17) + \frac{1}{n}\sum_{i=1}^{n} 2x_i^2 w_1$$

Split sum

$$0 = \sum_{i=1}^{n} -x_i (y_i - 17) + \sum_{i=1}^{n} x_i^2 w_1$$

$$\sum_{i=1}^{n} x_i^2 w_1 = \sum_{i=1}^{n} x_i (y_i - 17)$$

$$w_1 \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i (y_i - 17)$$

$$w_1 = \frac{\sum_{i=1}^{n} x_i (y_i - 17)}{\sum_{i=1}^{n} x_i^2}$$

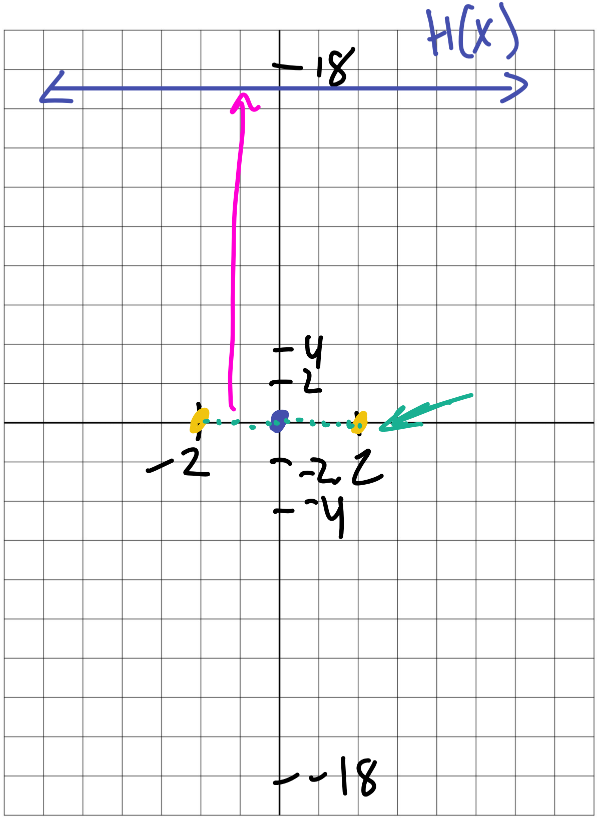
Prediction rule: $H(x) = 17 + w_1 x$

Problem 4.3

True or False: For an arbitrary dataset, the prediction rule $H^*(x) = 17 + w_1^* x$ goes through the point $(\bar{x}, \bar{y})$.

- True
- False

$$\frac{-2+2}{2}=0 \qquad \frac{0+0}{2}=0$$



Consider dataset $y = \{(-2, 0), (2, 0)\}$

$(\bar{x}, \bar{y})$ = $(0, 0)$

MSE

$$H(x) = w_0 + w_1 x$$

Prediction rule: $H^*(x) = 17 + w_1^* x$

$$w_1^* = \frac{\sum x_i (y_i - 17)}{\sum x_i^2} = \frac{34 + (-34)}{2^2 + 2^2 = 8} = 0$$

$$w_1^* = 0 \Rightarrow \text{slope} = 0$$

$$H^*(x) = 17 + 0 \cdot x = 17$$

$$H^*(x) = w_0 + w_1 x$$

Prediction rule: $H(x) = 17 + w_1 x$

Problem 4.4

regression line: $H(x) = w_0 + w_1 x$ that has the smallest MSE

True or False: For an arbitrary dataset, the mean squared error associated with $H^*(x)$ is greater than or equal to the mean squared error associated with the regression line.

- True
- False

$H^*(x) = 17 + 0x = 17$

$H(x) = 0$

03

Fall 2021 Midterm Exam

Problem 2

Problem 2

Source: Fall 2021 Midterm, Problem 2

Consider a set of 23 data points $y_1, y_2, y_3, ..., y_{23}$ such that $y_1 < y_2 < ... < y_{23}$. Let's call this Dataset A.

We create a new dataset, Dataset B, by repeating each point in Dataset A once. That is, Dataset B is the set of 46 points $y_1, y_1, y_2, y_2, ..., y_{23}, y_{23}$.

Answer the following questions regarding the relationship between Dataset A and Dataset B. **Justify your answers.**

Problem 2.1

Suppose the minimizer of mean absolute error $R_{abs}(h)$ for Dataset A is 5. What is the minimizer of mean absolute error for Dataset B? If you believe there are multiple minimizers, specify them all. If you believe you need more information to answer the question, state that clearly.

A $\quad y_1, \ldots, y_{23} \rightarrow 23$ pts $\rightarrow$ odd

median $= y_{12}$ exactly $= 5$

B $\quad y_1, \ldots, y_{46} \rightarrow 46$ pts even

SAME as A $\quad 5 = y_{23} \leq \text{median} \leq y_{24} = 5$

Problem 2.1

Suppose the minimizer of mean absolute error $R_{abs}(h)$ for Dataset A is 5. What is the minimizer of mean absolute error for Dataset B? If you believe there are multiple minimizers, specify them all. If you believe you need more information to answer the question, state that clearly.

2.1 For odd $n = 23$, the unique minimizer of R_{abs} is the median y_{12}, so $y_{12} = 5$. For even $n = 46$, the minimizers are $[b_{23}, b_{24}]$; by the key fact, $b_{23} = b_{24} = y_{12} = 5$. The interval collapses to a point:

$$\textbf{unique minimizer for Dataset B: } h = 5.$$

Problem 2.2

Suppose the *mean absolute deviation from the median* for Dataset A is 17. What is the *mean absolute deviation from the median* for Dataset B? If you believe you need more information to answer the question, state that clearly.

Problem 2.2

Suppose the *mean absolute deviation from the median* for Dataset A is 17. What is the *mean absolute deviation from the median* for Dataset B? If you believe you need more information to answer the question, state that clearly.

2.2 Let $m = 5$ be the shared median. For A: $\frac{1}{23}\sum|y_i - m| = 17 \Rightarrow \sum|y_i - m| = 391$. Each term appears twice in B, so

$$\text{mean abs. dev. for B} = \frac{2(391)}{46} = 17.$$

Doubling every point doubles both the numerator and n, so the mean is **unchanged: 17**.

Problem 2.3

Suppose the function $R_A(h)$ represents mean absolute error for Dataset A and $R_B(h)$ represents mean absolute error for Dataset B.

Is it true that $R_A(h) = R_B(h)$ for any real number h? (In other words, are the graphs of $R_A(h)$ and $R_B(h)$ identical?) Explain your reasoning.

$$\frac{1}{n} \sum |y_i - h|$$

Problem 2.3

Suppose the function $R_A(h)$ represents mean absolute error for Dataset A and $R_B(h)$ represents mean absolute error for Dataset B.

Is it true that $R_A(h) = R_B(h)$ for any real number h? (In other words, are the graphs of $R_A(h)$ and $R_B(h)$ identical?) Explain your reasoning.

2.3 Yes, $R_A(h) = R_B(h)$ for all h:

$$R_B(h) = \frac{1}{46}\sum_{j=1}^{46}|b_j - h| = \frac{1}{46}\cdot 2\sum_{i=1}^{23}|y_i - h| = R_A(h).$$

The graphs are **identical**, not just the minimizers. Duplication scales total loss and n by the same factor, leaving the average unchanged everywhere.

04

Advice and Recommendations

From Jordan and Eric

Advice

Breathe

Studies show that mindfulness can improve exam performance by freeing up brain space from test anxiety

Scavenger Hunt

The exam may or may not be in order of difficulty (which is subjective anyway), so start with what you know

Get Rest

Consistent sleep is an indicator of success.

Explainability

The easiest way to see that you've defended your proof properly is to imagine explaining it to someone with what you've written.

Don't be Hungry

Make sure to eat something before your exam. It'll give a good boost to your blood sugar and keep you energized. In fact, because you have a breakout room to yourself, keep a snack nearby!

Stay Hydrated

Keep a water bottle nearby! You never know when it'll come in handy.

THANKS!

DO YOU HAVE ANY QUESTIONS?

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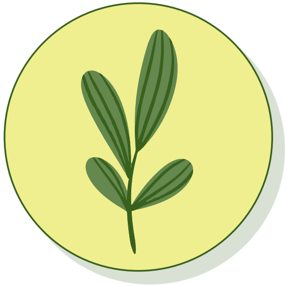
Here's an assortment of alternative resources whose style fits the one of this template:

VECTORS:

- Flat summer pattern design
- Flat tropical summer background with leaves
- Flat summer pattern design (II)



RESOURCES

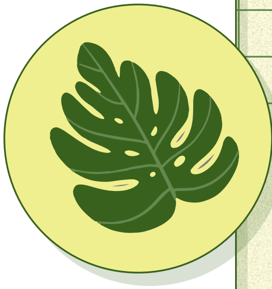
Did you like the resources on this template? Get them for free at our other websites:

PHOTOS:

- Top view delicious salad in an avocado
- Top view of avocado with asparagus and spinach
- Woman standing against wall drinking avocado smoothies in plastic cup
- African american man smiles

VECTORS:

- Flat summer pattern design
- Flat summer pattern design (II)
- Flat summer pattern design (III)
- Flat summer pattern design (IV)
- Flat tropical summer background with leaves



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